Activity 8A

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Given keys:

{4371, 1323, 6173, 4199, 4344, 9679, 1989}

Hash function:

$h(x) = x \bmod 10$

Key	h(x)
4371	1
1323	3
6173	3
4199	9
4344	4
~~[illegible]~~	
9679	9
1989	9

(a) Separate Chaining

Index	hashTable
0	–
1	4371
2	–
3	1323 → 6173
4	4344
5	–
6	–
7	–
8	–
9	4199 → 9679 → 1989

(b) Open addressing - linear probing

Formula: $h_i(x) = (h(x) + i) \bmod 10$

4371 → index 1
1323 → Index 3
6173 → 3 occupied → Index 4
4199 → Index 9
4344 → 4 occupied → Index 5
9679 → 9 occupied → 0
1989 → 9 occupied → 0 occupied → 1 occupied → 2

Final Table:

Index	value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	–
7	–
8	–
9	4199

(c) ... probing

Formula:

$h_i(x) = (h(x) + i^2) \bmod 10$

4371 → 1

1323 → 3

6173 → 3 occupied → $3+1^2 = 4$

4199 → 9

4344 → 4 occupied → 5

9679 → 9 occupied → 0

1989 → 9 occupied → 0 occupied → 3 occupied → 8

Final Table:

Index	Value
0	9679
1	4371
2	–
3	1323
4	6173
5	4344
6	–
7	–
8	1989
9	4199

(d) Open addressing - Double hashing

Given:

$h_2(x) = 7 - (x \bmod 7)$

The second hash values are:

Key	h(x)	$h_2(x)$
4371	1	4
1323	3	7
6173	3	1
4199	9	1
4344	4	3
9679	9	2
1989	9	6

Formula:

$h_i(x) = (h(x) + i \times h_2(x)) \bmod 10$

Insertion:

4371 → 1

1323 → 3

6173 → 3 occupied → 4

4199 → 9

4344 → 4 occupied → 7

9679 → 9 occupied → 1 occupied → 3 occupied

1989 → 9 → 5 → 1 → 7 → 3 → 9 ... all occupied

∴ 1989 cannot be inserted with the given second hash function because its step size is 6, which cycles only through the odd positions.

Final Table:

Index	Value
0	–
1	4371
2	–
3	1323
4	6173
5	9679
6	–
7	4344
8	–
9	4199

9 → 5 → 1 → 7 → 3 → 9 → ...

It keeps repeating because $h_2(1989) = 6$ and the table size

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